Linear Algebra I Recitation Class: Week 9

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the sets of rational, real, and complex numbers, respectively. V is a vector space over a field F. If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind. If T is a linear transformation, we use the notation $\ker T := N(T)$ and $\operatorname{im} T := R(T)$.

Let V be a finite dimensional vector space. The space of **linear functionals**

$$V^* := \operatorname{Hom}_F(V, F) = \{f : V \to F \mid f \text{ is linear}\}$$

is called the **dual space** of V; it is a vector space over F. If $\beta = \{v_1, v_2, \cdots, v_n\}$ is an ordered basis for V, for each v_i, we define $v_i^* \in V^*$ by $v_i^*(v_j) = \delta_{ij}$, where

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Then $\beta^* = \{v_1^*, \cdots, v_n^*\}$ is an ordered basis for V^*, which is called the **dual basis** of β, and we have an isomorphism:

$$\begin{aligned} V &\xrightarrow{\sim} V^* \\ a_1 v_1 + \cdots + a_n v_n &\longmapsto a_1 v_1^* + \cdots + a_n v_n^* \end{aligned}$$

Fixing the basis β for V, any element $v \in V$ can be seen as a column vector $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$. The isomorphism above tells us that any element $f \in V^*$ can be seen as a row vector $(b_1, \cdots, b_n)$ and that

$$f(v) = (b_1, \cdots, b_n) \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = b_1 a_1 + \cdots + b_n a_n$$

This isomorphism depends on the choice of basis. That is, if $\gamma = \{w_1, \cdots, w_n\}$ is another ordered basis for V, then we have a different isomorphism $V \xrightarrow{\sim} V^*$ sending $w_i \mapsto w_i^*$ for each i.

Remark. If V is an infinite dimensional vector space and F is an infinite field, then the dimension of V^* is strictly larger than the dimension of V. For example, let X be an infinite set and consider the space of functions with finite support

$$C_{\text{fin}}(X) := \{f : X \to F \mid f(x) = 0 \text{ for almost all } x \in X\}.$$

Then $C_{\text{fin}}(X)^*$ is isomorphic to the space of all functions $F^X := \{f : X \to F\}$. While $\dim C_{\text{fin}}(X) = |X|$, we have $\dim C_{\text{fin}}(X)^* = \dim F^X = |F|^{|X|} > \dim C_{\text{fin}}(X)$.

If W is another finite dimensional vector space, and $T : V \to W$ is linear, then for any linear functional $f \in W^*$, the composition $f \circ T : V \to W \to F$ is a linear functional in V^*. Therefore, we have a map (notice that the direction is reversed!):

$$T^* : W^* \longrightarrow V^* \qquad f \longmapsto T^*(f) = f \circ T.$$

Furthermore, T^* is linear. If we fix an ordered basis $\beta = \{v_1, \cdots, v_n\}$ for V, and an ordered basis $\gamma = \{w_1, \cdots, w_m\}$ for W, the matrix representation is $[T^*]_{\gamma^*}^{\beta^*} = \left([T]_\beta^\gamma\right)^t$.

For any $x \in V$ and $f \in V^*$, define $\widehat{x}(f) := f(x) \in F$. Then $\widehat{x} : V^* \to F$ is a linear functional, meaning $\widehat{x} \in (V^*)^* =: V^{**}$, which is called the **double dual space** of V. If V is finite dimensional, then we have an isomorphism:

$$\begin{aligned} V &\xrightarrow{\sim} V^{**} \\ x &\longmapsto \widehat{x} \end{aligned}$$

This isomorphism does NOT depend on the choice of basis, so we call such an isomorphism a **canonical isomorphism**, and we say V is **canonically isomorphic** to V^{**}.

Let V be a vector space, and S be a subset of V. The **annihilator** of S is the subset S^0 of V^* given by

$$S^0 := \{f \in V^* \mid f(s) = 0 \text{ for all } s \in S\}.$$

Note that:

- $0 \in S^0$, where 0 is the zero linear functional.
- For any $f, g \in S^0$ and $a \in F$, $(af + g)(s) = af(s) + g(s) = 0$ for all $s \in S$, so $af + g \in S^0$.

Hence S^0 is a vector subspace of V^*. One can easily see that $\{0\}^0 = V^*$, $V^0 = \{0\}$, and $S \neq \{0\} \implies S^0 \neq V^*$. Furthermore, if $R \subseteq S$, then for any $f \in S^0$, $f(r) = 0$ for all $r \in R \subseteq S$, thus $S^0 \subseteq R^0$. That is, the operator $(-)^0$ is inclusion-reversing.

We have some interesting properties regarding the annihilator:

Proposition 1. *Let V be a finite dimensional vector space over F. Then*

1. *If W is a subspace of V, then $W^* \cong V^*/W^0$. In particular,*

$$\dim W^0 = \dim V - \dim W$$

2. *If W is a subspace of V, then $(V/W)^* \cong W^0$.*
3. *If S is a subset of V, then $(S^0)^0 = \operatorname{span}(\widehat{S})$, where $\widehat{S} := \{\widehat{s} \in V^{**} \mid s \in S\}$. In particular, if W is a vector subspace of V, then $(W^0)^0 = \widehat{W}$.*
4. *If W_1, W_2 are subspaces of V, then $W_1^0 = W_2^0 \iff W_1 = W_2$.*
5. *If W_1, W_2 are subspaces of V, then $(W_1 + W_2)^0 = W_1^0 \cap W_2^0$.*

Proof. 1. Note that we have the inclusion map $\iota : W \to V$ given by $\iota(w) = w$. It is linear and induces the linear map $\iota^* : V^* \to W^*$ via $f \mapsto f \circ \iota = f|_W$. We claim that ι^* is onto: given $g \in W^*$, we want to find $f \in V^*$ such that $f|_W = g$. Let $\gamma = \{w_1, \cdots, w_n\}$ be a basis for W, which extends to a basis $\gamma \cup \{v_1, \cdots, v_m\}$ for V. Then we define

$$f(w_i) = g(w_i) \text{ for any } i = 1, \cdots, n \text{ and } f(v_j) = 0 \text{ for any } j = 1, \cdots, m$$

and extend it linearly. One can see that $f \in V^*$ and $f|_W = g$. Furthermore,

$$\begin{aligned} \ker \iota^* &= \{f \in V^* \mid f|_W = 0\} \\ &= \{f \in V^* \mid f(w) = 0 \text{ for all } w \in W\} = W^0. \end{aligned}$$

Thus, by the first isomorphism theorem, $V^*/W^0 \cong W^*$. Taking dimensions,

$$\dim W = \dim W^* = \dim(V^*/W^0) = \dim V^* - \dim W^0 = \dim V - \dim W^0.$$

2. For any element $f \in W^0$, define a map $\overline{f} : V/W \to F$ by $\overline{f}(v + W) = f(v)$. $\overline{f}$ is well-defined: if $v + W = v' + W$, then $v - v' \in W$, so $f(v) - f(v') = f(v - v') = 0$ since $f \in W^0$. One can easily see that $\overline{f}$ is linear, thus $\overline{f} \in (V/W)^*$. Therefore, we can define a map

$$\psi : W^0 \to (V/W)^* \qquad f \mapsto \overline{f}.$$

One can check that ψ is linear. Furthermore, given $f \in W^0$ such that $\overline{f} = 0$, we have $\overline{f}(v+W) = f(v) = 0$ for all $v \in V$. In particular, if $\gamma = \{w_1, \cdots, w_m\}$ is a basis for W which extends to a basis $\beta = \gamma \cup \{v_1, \cdots, v_n\}$ for V, then $\{v_1 + W, \cdots, v_n + W\}$ is a basis for V/W. We have $\overline{f}(v_i + W) = f(v_i) = 0$ for each $i = 1, \cdots, n$, and $f(w_j) = 0$ for all $j = 1, \cdots, m$ since $f \in W^0$. Consequently, $f = 0$, meaning ψ is one-to-one. Note that $\dim(V/W)^* = \dim(V/W) = \dim V - \dim W = \dim W^0$, thus ψ is an isomorphism.

3. For any $\hat{x} \in \operatorname{span}(\widehat{S})$, $\hat{x} = a_1\widehat{s_1} + \cdots + a_n\widehat{s_n}$ for some $a_i \in F$ and $s_i \in S$. Then for any $f \in S^0$,

$$\hat{x}(f) = (a_1\widehat{s_1} + \cdots + a_n\widehat{s_n})(f) = a_1 f(s_1) + \cdots + a_n f(s_n) = 0.$$

Therefore, $\operatorname{span}(\widehat{S}) \subseteq (S^0)^0$. To show equality, let $W = \operatorname{span}(S)$. We claim that $W^0 = S^0$. It is clear that $W^0 \subseteq S^0$ since $S \subseteq W$. On the other hand, for any $f \in S^0$, $f(w) = f(\sum a_i s_i) = \sum a_i f(s_i) = 0$ for any $w \in W$, thus $W^0 = S^0$. By the dimension formula above, $\dim(S^0)^0 = \dim V^* - \dim S^0 = \dim V - \dim W^0 = \dim W = \dim \operatorname{span}(\widehat{S})$ (since $\widehat{-}$ is an isomorphism). Therefore, $(S^0)^0 = \operatorname{span}(\widehat{S})$.

4. If $W_1 = W_2$, clearly $W_1^0 = W_2^0$. Conversely, if $W_1^0 = W_2^0$, then $(W_1^0)^0 = (W_2^0)^0$. This implies $\widehat{W_1} = \widehat{W_2}$, and therefore $W_1 = W_2$ since $\widehat{-}$ is an isomorphism.

5. For $f \in (W_1 + W_2)^0$, $f(w_1) = 0$ for any $w_1 \in W_1$ and $f(w_2) = 0$ for any $w_2 \in W_2$; therefore, $f \in W_1^0 \cap W_2^0$. On the other hand, for $f \in W_1^0 \cap W_2^0$, given $w = w_1 + w_2 \in W_1 + W_2$, $f(w) = f(w_1 + w_2) = f(w_1) + f(w_2) = 0$, hence $f \in (W_1 + W_2)^0$.

□

Exercise A. In the same setting as above, prove that $(W_1 \cap W_2)^0 = W_1^0 + W_2^0$.

With the notion of the annihilator, we can investigate some interesting facts about the transpose $T^* : W^* \to V^*$ of any linear transformation $T : V \to W$.

Proposition 2. *Let V, W be finite dimensional vector spaces over F and $T : V \to W$ be linear, which induces $T^* : W^* \to V^*$:*

1. $\ker T^* = (\operatorname{im} T)^0$.
2. $(\ker T)^0 = \operatorname{im} T^*$.

Proof. 1. Given $f \in \ker T^* \subseteq W^*$, then $T^*(f) = f \circ T = 0$, that is, $f(T(v)) = 0$ for all $v \in V$, i.e., $f(w) = 0$ for all $w \in \operatorname{im} T$, thus $f \in (\operatorname{im} T)^0$. Conversely, given $g \in (\operatorname{im} T)^0$, $T^*(g)(v) = g(T(v)) = 0$ for all $v \in V$, thus $g \in \ker T^*$.

2. The proof is similar to the one above: Given $f \in \operatorname{im} T^*$, then $f = T^*(g) = g \circ T$ for some $g \in W^*$. Given $x \in \ker T$, $f(x) = g(T(x)) = g(0) = 0$; therefore, $f \in (\ker T)^0$. By the dimension formula,

$$\begin{aligned} \dim \operatorname{im} T^* &= \dim W^* - \dim \ker T^* \\ &= \dim W^* - \dim(\operatorname{im} T)^0 \\ \text{Proposition 1} &= \dim W^* - (\dim W - \dim \operatorname{im} T) = \dim \operatorname{im} T \end{aligned}$$

On the other hand,

$$\begin{aligned} \dim(\ker T)^0 &= \dim V^* - \dim \ker T \\ &= \dim V^* - (\dim V - \dim \operatorname{im} T) = \dim \operatorname{im} T = \dim \operatorname{im} T^* \end{aligned}$$

Thus the two spaces coincide.

□

There are two important consequences of Proposition 2:

Corollary 3. *Let V, W be finite dimensional vector spaces over F and $T : V \to W$ be linear.*

1. *T^* is one-to-one if and only if T is onto.*
2. *T^* is onto if and only if T is one-to-one.*

Equivalently, given a short exact sequence of finite dimensional vector spaces:

$$0 \longrightarrow V_1 \xrightarrow{T_1} V_2 \xrightarrow{T_2} V_3 \longrightarrow 0$$

We have (be careful of the direction of the arrows):

$$0 \longleftarrow V_1^* \xleftarrow{T_1^*} V_2^* \xleftarrow{T_2^*} V_3^* \longleftarrow 0$$

In the language of category theory, we say that $(-)^* = \operatorname{Hom}_F(-, F)$ is a **contravariant exact functor** $(-)^* : \mathrm{FdVect}_F \to \mathrm{FdVect}_F^{\mathrm{op}}$.

Corollary 4. *Let $A \in M_{m\times n}(F)$, then* $\operatorname{rank} L_A = \operatorname{rank} L_{A^t}$.

Proof. Let β be the standard ordered basis for F^n and γ be the standard basis for F^m. Recall that $L_A : F^n \to F^m$ by $v \mapsto Av$ and $[L_A]^\gamma_\beta = A$, which induces

$$L_A^* : (F^m)^* \to (F^n)^* \quad f \mapsto f \circ L_A.$$

and $[L_A^*]^{\beta^*}_{\gamma^*} = A^t$. Therefore, $\operatorname{rank} L_A^* = \operatorname{rank} L_{A^t}$. It suffices to show that $\operatorname{rank} L_A^* = \operatorname{rank} L_A$. By the dimension formula:

$$\begin{aligned}
\operatorname{rank} L_A^* &= m - \dim \ker L_A^* \\
&\overset{\text{Proposition 2}}{=} m - \dim(\operatorname{im} L_A)^0 \\
&\overset{\text{Proposition 1}}{=} m - (m - \dim \operatorname{im} L_A) = \operatorname{rank} L_A.
\end{aligned}$$

□

Recall that V^{**} is canonically isomorphic to V when V is finite dimensional, so this provides motivation to investigate the double transpose $T^{**} := (T^*)^*$ of a linear transformation $T : V \to W$, where W is also finite dimensional. It turns out that T^{**} is essentially T via the identification $v \mapsto \widehat{v}$.

Proposition 5. *Let V, W be finite dimensional vector spaces over F, and $T : V \to W$ be linear. T induces a linear map $T^* : W^* \to V^*$, and T^* induces a linear map $T^{**} : V^{**} \to W^{**}$. In this case, we have a commutative diagram:*

$$\begin{array}{ccc}
V & \xrightarrow{T} & W \\
\widehat{\ }\downarrow\wr & & \wr\downarrow\widehat{\ } \\
V^{**} & \xrightarrow[T^{**}]{} & W^{**}
\end{array}$$

Proof. It suffices to prove that $T^{**}(\hat{v}) = \widehat{T(v)}$ for any $v \in V$. Since $T^{**}(\hat{v}), \widehat{T(v)} \in W^{**}$, both are linear functionals on W^*. For any $f \in W^*$, compute

$$\begin{aligned}
T^{**}(\hat{v})(f) &= \hat{v} \circ T^*(f) \\
&= \widehat{v} \circ (f \circ T) \\
&= (f \circ T)(v) = f(T(v)) = \widehat{T(v)}(f).
\end{aligned}$$

Therefore, $T^{**}(\hat{v}) = \widehat{T(v)}$. □

Exercise B. In the same setting as above, prove that $\widehat{\ker T} = \ker T^{**}$ and that $\widehat{\operatorname{im} T} = \operatorname{im} T^{**}$. (Hint: The commutative diagram above is useful.)

To conclude, we have the following duality:

1. Subspaces are dual to quotient spaces:

$$\begin{array}{ccccc}
V & & V^* = \operatorname{Hom}_F(V, F) & & V^{**} \cong V \\
\text{subspace } W & \rightsquigarrow & \text{quotient space } V^*/W^0 & \rightsquigarrow & \text{subspace } W^{00} = \widehat{W} \cong W \\
\text{quotient space } V/W & \rightsquigarrow & \text{subspace } W^0 & \rightsquigarrow & \text{quotient space } V^{**}/W^{00} \cong V/W
\end{array}$$

2. "De Morgan's laws" for the annihilator of subspaces: Let W_1, W_2 be subspaces of V,
$$(W_1 \cap W_2)^0 = W_1^0 + W_2^0 \qquad (W_1 + W_2)^0 = W_1^0 \cap W_2^0$$

3. Injectivity is dual to surjectivity:
$$T : V \to W \qquad T^* : W^* \to V^* \qquad T^{**} : V^{**} \to W^{**}$$
$$T \text{ is one-to-one} \rightsquigarrow T^* \text{ is onto} \rightsquigarrow T^{**} \text{ is one-to-one } (T \text{ is one-to-one})$$
$$T \text{ is onto} \rightsquigarrow T^* \text{ is one-to-one} \rightsquigarrow T^{**} \text{ is onto } (T \text{ is onto})$$

4. Kernels are dual to cokernels; Images are dual to coimages:
$$T : V \to W \qquad T^* : W^* \to V^* \qquad T^{**} : V^{**} \to W^{**}$$
$$\operatorname{im} T \rightsquigarrow \begin{array}{c}(\operatorname{im} T)^* \cong W^*/(\operatorname{im} T)^0 \\ = W^*/\ker T^* =: \operatorname{coim} T^*\end{array} \rightsquigarrow (W^*/(\operatorname{im} T)^0)^* \cong \widehat{\operatorname{im} T} = \operatorname{im} T^{**}$$
$$\ker T \rightsquigarrow \begin{array}{c}(\ker T)^* \cong V^*/(\ker T)^0 \\ = V^*/\operatorname{im} T^* =: \operatorname{coker} T^*\end{array} \rightsquigarrow (V^*/(\ker T)^0)^* \cong \widehat{\ker T} = \ker T^{**}$$

The notion of duality also appears in different algebraic structures. For example, in finite groups, the dual of G is a set consisting of *characters* of G, that is:
$$\widehat{G} := \{\chi : G \to \mathbb{C} \mid \chi(gh) = \chi(g)\chi(h)\}.$$
For a special kind of group G, which is called an LCA[1] group, the *Pontryagin dual* $\widehat{G}$ of G consists of continuous *unitary character*, that is:
$$\widehat{G} := \{\chi : G \xrightarrow{\text{conti.}} S^1 \mid \chi(gh) = \chi(g)\chi(h)\} \qquad S^1 = \{z \in \mathbb{C} \mid |z| = 1\}.$$
The pontryagin dual of LCA groups plays an important role in abstract Fourier transformation and number theory.

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